

- 1 (a) length, current, temperature, amount of substance, (luminous intensity)
any three, 1 each B3 [3]
- (b) (i) $F: \text{kg m s}^{-2}$ B1
 $\rho: \text{kg m}^{-3}$ B1
 $v: \text{m s}^{-1}$ B1 [3]
- (ii) some working e.g. $\text{kg m s}^{-2} = \text{m}^2 \text{kg m}^{-3} (\text{m s}^{-1})^k$ M1
hence $k = 2$ A1 [2]
- 2 (a) (i) horizontal speed constant at 8.2 m s^{-1} C1